Sahaj Bhadja

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Summary

Full-stack engineer with 2+ years building **Python backends**, **React/TypeScript interfaces**, and **scalable distributed systems** across AWS-backed platforms, with experience shipping user-facing products in fast-paced environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **Python-first platform services** for 500+ users, reducing request latency from **1.2s to 350ms** by optimizing execution paths, tuning PostgreSQL queries, and improving responsiveness for user-facing analytics dashboards.
- Implemented **automated validation and ETL pipelines** for 5,000+ monthly jobs, reducing downstream failures and accelerating incident triage through earlier detection of data-quality drift and runtime anomalies.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **full-stack features and REST APIs** handling 10,000+ daily requests, translating product requirements into performant backend services and responsive React interfaces while shipping rapidly in a startup environment.
- Built **AI-assisted grading workflows** with multimodal models, reducing feedback turnaround by **60%** while iterating quickly based on user feedback to deliver delightful product experiences.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **large-scale experimentation pipelines** across 100,000+ records, enabling reproducible algorithm evaluation and data-driven decision-making while improving signal quality for model-driven workflows.
- Produced **technical reports and test scenarios** translating findings into actionable guidance, improving cross-team alignment and strengthening engineering quality as prototypes matured toward production.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations and stable delivery across production workloads at scale.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and improving confidence in frequent releases through rigorous code review practices.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **MERN-based platform** with WebSockets and RESTful APIs, syncing 2M+ records across locations with **33% fewer sync conflicts** and real-time updates for performant user-facing dashboards.
- Automated **Docker-based build and deployment flows** with Jenkins, cutting release time by **50%** and improving deployment consistency while holding code to high quality standards.

Skills

Languages: **Python**, **TypeScript**, JavaScript, Java, C++, SQL, Bash
Frontend: **React**, Next.js, TailwindCSS, component-driven design, responsive UI
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets
Databases: **PostgreSQL**, MySQL, MongoDB, Redis, relational schema design, query tuning
Cloud & Delivery: **AWS** (EC2, S3, Glue, ECS), **Docker**, Kubernetes, CI/CD, Git

Projects

FinNexus Live | *Next.js, TypeScript, Plaid, MongoDB, WebSockets, Kubernetes*

- Built a **scalable fintech platform** supporting 50,000+ users and 1M+ transactions, with Plaid bank linking, real-time WebSocket updates, and Kubernetes-based deployment for a performant multi-sided marketplace.

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*

- Built an **AI-powered Next.js SaaS platform** for image editing and search, integrating Stripe billing and MongoDB-backed workflows while improving premium conversion by **18%** through polished user experiences.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, Convex, Clerk*

- Built an **AI-powered document platform** with semantic search and real-time collaboration, supporting 200+ users with 95% query accuracy and reducing document review time by **60%** through intuitive UX.